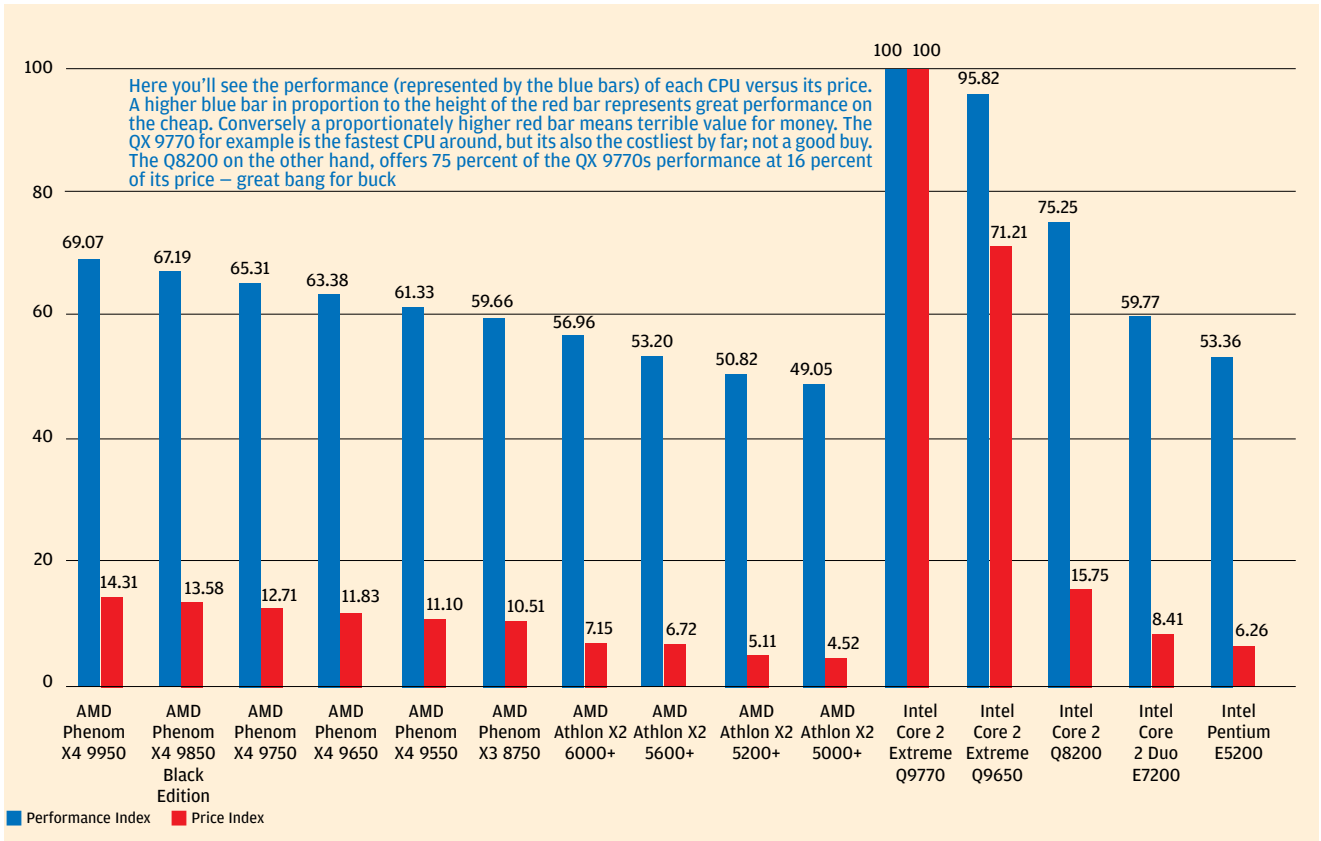




CPUs															
Brand Model	AMD Phenom X4 9950	AMD Phenom X4 9850 Black Edition	AMD Phenom X4 9750	AMD Phenom X4 9650	AMD Phenom X4 9550	AMD Phenom X3 8750	AMD Athlon X2 6000+	AMD Athlon X2 5600+	AMD Athlon X2 5200+	AMD Athlon X2 5000+	Intel Core 2 Extreme Q9770	Intel Core 2 Extreme Q9650	Intel Core 2 Q8200	Intel Core 2 Duo E7200	Intel Pentium E5200
Price (Rs)	9800	9300	8700	8100	7600	7200	4900	4600	3500	3100	68450	48750	10785	5760	4285
Performance Scores (Out of 100)	63.13	61.41	59.69	57.93	56.05	54.52	52.06	48.63	46.45	44.84	91.39	87.58	68.77	54.63	48.77
Features															
Core Frequency	2600 MHz	2500 MHz	2400 MHz	2300 MHz	2200 MHz	2400 MHz	3100 MHz	2900 MHz	2700 MHz	2600 MHz	3200 MHz	3000 MHz	2.33 GHz	2.53 GHz	2.5 GHz
Number Of Cores	4	4	4	4	4	3	2	2	2	2	4	4	4	2	2
External Bus Frequency	2000 MHz	2000 MHz	1800 MHz	1800 MHz	1800 MHz	1800 MHz	1000 MHz	1000 MHz	1000 MHz	1000 MHz	1600 MHz	1333 MHz	1333 MHz	1066 MHz	800 MHz
Fabrication Process (in Nanometers)	65 nm	65 nm	65 nm	65 nm	65 nm	65 nm	65 nm	65 nm	65 nm	65 nm	45 nm	45 nm	45 nm	45 nm	45 nm
L1 Cache	512 KB	512 KB	512 KB	512 KB	512 KB	384 KB	256 KB	256 KB	256 KB	256 KB	256 KB	256 KB	256 KB	128 KB	128 KB
L2 Cache	4 x 512 KB	4 x 512 KB	4 x 512 KB	4 x 512 KB	4 x 512 KB	3 x 512 KB	2 x 512 KB	2 x 512 KB	2 x 512 KB	2 x 512 KB	2 x 6144 KB	2 x 6144 KB	2 x 2048 KB	3072 KB	2048 KB
L3 Cache (if any)	2 MB	2 MB	2 MB	2 MB	2 MB	2 MB	N	N	N	N	N	N	N	N	N
Performance															
Synthetic Benchmarks															
PC Mark05 (CPU Score)	7713	7410	7233	6829	6533	6462	6325	5851	5526	5313	10321	9674	7540	6498	6141
3D Mark 2006 (CPU Score)	3886	3812	3624	3493	3358	2912	2375	2221	2075	2005	5238	4984	3791	2350	2200
Wprime (32 Million) #	15.125 sec	15.719 sec	16.375 sec	17.062 sec	17.859 sec	21.953 sec	27.64 sec	29.578 sec	31.733 sec	32.953 sec	13.296 sec	14.156 sec	18.203 sec	33.734 sec	35.5 sec
ScienceMark 2.0															
Cryptography (AES Encryption, Rijndael) #	10.059 sec	10.476 sec	10.86 sec	11.325 sec	11.805 sec	10.864 sec	8.54 sec	9.121 sec	9.8 sec	10.218 sec	8.933 sec	9.61 sec	12.205 sec	11.242 sec	12.052 sec
Membench (Memory Bandwidth)	7544.98	7590	7209	7234	7116.1	7161.26	8548.86	7709	8350.19	7894.32	7699.68	6885.15	6964.28	5767.34	4545.6
CineBench R10 (CPU Rendering Test 1)	8026	7785	7744	7399	7105	5839	4959	4625	4366	4207	12579	11843	8866	5326	4912
PovRay 3.7 (Chess 2, Real Time) #	4798 sec	48.55 sec	50.38 sec	52.61 sec	55.26 sec	67.9 sec	88.95 sec	94.23 sec	101.27 sec	105.41 sec	38.48 sec	41.2 sec	52.73 sec	95.11 sec	100.5 sec
Real World Benchmarks															
Rendering:															
Maya 8.0 (Gobalillum.ma, 640 x 480, Prod. Quality) #	37 sec	38 sec	39 sec	40 sec	41 sec	36 sec	39 sec	42 sec	45 sec	47 sec	20.9 sec	21 sec	21 sec	34.5 sec	36.9 sec
Compression:															
WinRAR 3.8 (KBps)	1836	1813	1793	1762	1742	1521	1255	1180	1182	1137	2156	2188	1691	1238	1043
Encoding:															
DivX Encode 6.8.0.19 (sec) #	149 sec	157 sec	161 sec	167 sec	177 sec	168 sec	148 sec	157 sec	171 sec	172 sec	107.1 sec	113.2 sec	145.7 sec	136.4 sec	148 sec
Gaming:															
UT 3 (800 x 600, All Visual Settings Off)	109	108	103	102	101	103	103	97	95	92	216	213	142	151	129
Doom 3 (640 x 480, Low Detail)	196.3	193.4	183.2	179.3	177.8	188.2	182	173.4	166.8	161.6	311.3	299.4	209.9	219.9	184.4
Company Of Heroes (800 x 600, Min Detail)	240.7	225	247	252.2	243.9	265.8	240.9	221	217.2	208.5	570.2	542.7	285.8	333.3	250.9
# Lower Is Better															



the planet (for now). The QX 9770 is clearly a super high-end part. It features an FSB speed of 1600 MHz and an unlocked multiplier – overclockers rejoice. The bad – at Rs 68,450, you can actually build an entire PC with a GTX 260 graphics card for the price. At Rs 48,750, the QX 9650 is cheaper, but not affordably so; you could assemble a good gaming PC for the price. Ideally these CPUs will be sought after by only those who can afford to pay for them. Our Best Performance award goes to Intel’s QX 9770, the fastest CPU across the board; albeit the costliest. Despite some brave and rather heroic efforts from the likes of the Phenom X4 9950 and the X4 9850 Black Edition, Intel’s Q 8200 gets our Best Buy award. At Rs 10,785, it is fast enough to warrant you spending that much dough on it.

For our Editors Pick we were looking at something a little more affordable. At the same time we wanted something rea-

sonably fast and future proof. All this pointed towards a quad-core, but which one? At just Rs 400 more than the Phenom X3 8750 (a tri-core), AMDs Phenom X4 9550 is a bargain; its price – Rs 7,600. It will satisfy almost everyone but the demanding gamer or 3D content creator. This is one processor that has bang for buck written all over it. In case you can spend no more than Rs 6,000 on a CPU we feel Intel’s E7200 is your best bet. For those looking at a basic multimedia cum home PC on a shoestring, AMDs 5000+ is hard to beat – it’ll even play HD content easily.

What’s Next?

Parallelism, in bucketfuls; Nehalem is supposed to be scalable so expect to see even eight core versions in 2009. Optionally Intel could drop in a G45 graphic core in place of a single core and give you a fast seven core processor with inbuilt graphic

solution. Of course cheaper quad cores will emerge as well, as dual cores slowly become obsolete. Besides the X58 we should see new LGA 1366 platforms from NVIDIA. The fourth quarter will see the emergence of *Westmere*; a 32 nm version of the Nehalem architecture. Later, in 2010 we’ll see a new architecture in the form of *Sandy Bridge*. Then there is *Larrabee* lurking somewhere within Intel’s deepest labs; but Intel is close mouthed about their new GPU killer.

AMD isn’t hatching eggs either. After their Phenom X4s bombed badly they’re all set to pull up their trousers with *Deneb*. Deneb is 45 nm, and supposed to be 20 per cent faster than Intel’s Kentsfield; interesting eh? A pity they didn’t launch it earlier. Besides Kentsfield, Deneb now also has to contend with the shiny newcomer which is by far, the fastest desktop CPU we’ve ever tested. ■

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